US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12397693
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Salter; Stuart C. et al.

Vehicle console assembly

Abstract

A vehicle console system includes a body having first and second sidewalls. Each of the first and second sidewalls defines a rail along an inner surface thereof. A support bar slidably engages the rails. The support bar defines a track. A cover assembly is operably coupled to the support bar. The cover assembly includes a base panel coupled to the support bar via a connector assembly and a deployable panel rotatably coupled to the base panel. The cover assembly is configured to translate along a first movement path with the support bar along the rails and a second movement path along the track of the support bar. The cover assembly is configured to rotate along a third movement path between a first side position and a second side position.

Inventors: **Salter; Stuart C. (White Lake, MI), Shedlarski; Dustin (Commerce Township, MI), Welch; Ryan (Ottawa Lake, MI), Dellock; Paul Kenneth (Northville, MI), Huebner; Annette Lynn (Highland, MI), Johnson; Aaron (Allen Park, MI)**

Applicant: **Ford Global Technologies, LLC** (Dearborn, MI)

Family ID: **1000008780258**

Assignee: **Ford Global Technologies, LLC (Dearborn, MI)**

Appl. No.: **18/079955**

Filed: **December 13, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240190321 A1	Jun. 13, 2024

Publication Classification

Int. Cl.: **B60N3/00** (20060101); **B60N2/75** (20180101); **B60Q3/225** (20170101); **B60R7/04** (20060101); **B60R11/02** (20060101); **B60R11/04** (20060101); B60R11/00 (20060101)

U.S. Cl.:

CPC **B60N3/002** (20130101); **B60N2/79** (20180201); **B60Q3/225** (20170201); **B60R7/04** (20130101); **B60R11/0229** (20130101); **B60R11/04** (20130101); B60R2011/0007 (20130101)

Field of Classification Search

CPC: B60N (3/001); B60N (3/002); B60N (2/79); B60N (2/793); B60N (2/773); B60N (2/777); B60R (7/04); B60R (2011/0007); A47B (5/006); B60Q (3/225); B60Q (3/233); B64D (11/063)

USPC: 296/37.8; 296/24.34; 108/44; 701/36; 297/145

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5547247	12/1995	Dixon	N/A	N/A
6032587	12/1999	Salenbauch	108/44	B60N 3/002
6220660	12/2000	Bedro	297/146	B60N 2/79
6793281	12/2003	Duerr	297/173	B60N 2/753
7293507	12/2006	DePue et al.	N/A	N/A
7798072	12/2009	Becker	297/145	A47C 7/70
8540297	12/2012	Browne et al.	N/A	N/A
9896011	12/2017	Kong	N/A	B60N 3/002
10457170	12/2018	Line et al.	N/A	N/A
10457171	12/2018	Line et al.	N/A	N/A
10919426	12/2020	Campagnone	N/A	B60R 7/04
10926880	12/2020	Mayne	N/A	A47B 31/06
12043155	12/2023	Thomas	N/A	B60N 2/77
2010/0201147	12/2009	Jones	296/24.34	B60N 2/777
2010/0244478	12/2009	DePue	296/37.8	B60N 2/773
2013/0008930	12/2012	Hipshier et al.	N/A	N/A
2014/0207333	12/2013	Vandivier	701/36	B60N 2/0278
2018/0072209	12/2017	Farooq et al.	N/A	N/A
2020/0055436	12/2019	Garcia Sanchez	N/A	B60R 21/026
2023/0150400	12/2022	Kim	701/49	B60N 2/143
2024/0278701	12/2023	Stout, II	N/A	B60N 2/79

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
114987309	12/2021	CN	B60N 3/001
10258305	12/2003	DE	B60N 2/468
102012015039	12/2013	DE	B60Q 3/74
102019120974	12/2020	DE	N/A
102019130142	12/2020	DE	N/A
2005193855	12/2004	JP	N/A
5715268	12/2014	JP	N/A
2012174613	12/2011	WO	N/A

OTHER PUBLICATIONS

Mueller et al. (DE 102012015039 A1), machine translation (Year: 2014). cited by examiner
Wang et al. (CN 114987309 A), machine translation (Year: 2022). cited by examiner

---

Primary Examiner: Weisberg; Amy R

Assistant Examiner: Zhuo; Wenwei

Attorney, Agent or Firm: Price Heneveld LLP

Background/Summary

FIELD OF THE DISCLOSURE
(1) The present disclosure generally relates to a console assembly. More specifically, the present disclosure relates to a console assembly for a vehicle.
BACKGROUND OF THE DISCLOSURE
(2) Many vehicles include center consoles. The center consoles often provide storage for housing items. Additionally, the center consoles often have a padded layer on a top thereof.
SUMMARY OF THE DISCLOSURE
(3) According to at least one aspect of the present disclosure, a vehicle console system includes a body having first and second sidewalls. Each of the first and second sidewalls defines a rail along an inner surface thereof. A support bar slidably engages the rails. The support bar defines a track. A cover assembly is operably coupled to the support bar. The cover assembly includes a base panel coupled to the support bar via a connector assembly and a deployable panel rotatably coupled to the base panel. The cover assembly is configured to translate along a first movement path with the support bar along the rails and a second movement path along the track of the support bar. The cover assembly is configured to rotate along a third movement path between a first side position and a second side position.
(4) According to another aspect of the present disclosure, a vehicle console assembly includes a body defining a storage cavity between first and second sidewalls. A support bar is coupled to each of the first and second sidewalls. The support bar defines a track extending between the first and second sidewalls. A cover assembly is operably coupled to the support bar. The cover assembly includes a first panel. A second panel is rotatably coupled to the first panel. The second panel is operable between a folded state and a deployed state. A connector assembly is coupled to a lower surface of the first panel and extends through the track of the support bar. The cover assembly is configured to rotate in a first direction from a center position to a first side position and in a second direction from the center position to a second side position. The cover assembly is configured to translate along the track.
(5) According to another aspect of the present disclosure, a console assembly for a vehicle includes a body having sidewalls. Each sidewall defines a rail. A support bar slidably engages the rails. The support bar defines a track. A cover assembly is operably coupled to the support bar. A connector assembly is coupled to the cover assembly and engages the support bar. The cover assembly is configured to translate along a first movement path with the support bar via the rails, translate along a second movement path along the track via the connector assembly, and rotate about a vertical axis along a third movement path via the connector assembly.
(6) These and other aspects, objects, and features of the present disclosure will be understood and

appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following is a description of the figures in the accompanying drawings. The figures are not necessarily to scale, and certain features and certain views of the figures may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.
(2) In the drawings:
(3) FIG. **1** is a side perspective view of a console assembly within an interior compartment of a vehicle, according to the present disclosure;
(4) FIG. **2** is a side perspective view of a console assembly with a movable cover assembly in phantom, according to the present disclosure;
(5) FIG. **3** is a cross-sectional view of a console assembly with a movable cover assembly, according to the present disclosure;
(6) FIG. **4** is a side perspective view of a console assembly with a cover assembly moved along a first movement path, according to the present disclosure;
(7) FIG. **5** is a side perspective view of a console assembly with a cover assembly in a first side position and a deployed state, according to the present disclosure;
(8) FIG. **6** is a side perspective view of a console assembly with a cover assembly in a second side position and a deployed state, according to the present disclosure;
(9) FIG. **7** is a side perspective view of a console assembly with a cover assembly in a second side position and a deployed state moved along a second movement path, according to the present disclosure;
(10) FIG. **8** is a top perspective view of a console assembly with a cover assembly having an elongated panel rotatable along a third movement path, according to the present disclosure;
(11) FIG. **9** is a top perspective view of a console assembly with a cover assembly having an elongated panel moved along a second movement path, according to the present disclosure;
(12) FIG. **10** is a side perspective view of a console assembly with a three-panel cover assembly in a deployed state, according to the present disclosure; and
(13) FIG. **11** is a block diagram of a console system for a vehicle, according to the present disclosure.
DETAILED DESCRIPTION
(14) Additional features and advantages of the presently disclosed device will be set forth in the detailed description which follows and will be apparent to those skilled in the art from the description, or recognized by practicing the device as described in the following description, together with the claims and appended drawings.
(15) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. **1**. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.
(16) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A,

B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.
(17) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.
(18) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.
(19) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.
(20) Referring to FIGS. **1-11**, reference numeral **10** generally designates a console system **10** for a vehicle **12** which includes a console assembly **14**. The console assembly **14** includes a body **16** having first and second sidewalls **18**, **20** where each of the first and second sidewalls **18**, **20** defines a rail **22**, **24** on an inner surface **26** thereof. A support bar **28** slidably engages the rails **22**, **24** and defines a track **30**. A cover assembly **32** is operably coupled to the support bar **28**. The cover assembly **32** includes a base panel **34** coupled to the support bar **28** via a connector assembly **36** and a deployable panel **38** rotatably coupled to the base panel **34**. The cover assembly **32** is configured to translate along a first movement path **40** with the support bar **28** along the rails **22**, **24** and a second movement path **42** along the track **30** of the support bar **28**. The cover assembly **32** is also configured to translate along a third movement path **44**, rotating in a first direction to a first side position and in a second direction to a second side position.
(21) With reference to FIG. **1**, the vehicle **12** includes an interior compartment **60** with at least one seating row **62**, including a first seating assembly **64** and a second seating assembly **66**. Any practicable number of seating rows **62** may be included in the vehicle **12** without departing from the teachings herein. The illustrated seating row **62** is a front seating row **62** with the first seating assembly **64** being a driver seat **64** and the second seating assembly **66** being a passenger seat **66**. In the illustrated example, the driver seat **64** and the passenger seat **66** are disposed proximate to a dashboard having a center stack **70** and are separated by the console assembly **14**.
(22) Referring still to FIG. **1**, as well as FIG. **2**, the console assembly **14** includes the body **16** having a first end **80**, which is generally a vehicle-rearward end, and a second end **82**, which is generally a vehicle-forward end. The body **16** includes a lower base **84** configured to engage a vehicle body and an upper rim **86** coupled to the lower base **84**. In the illustrated configuration, the upper rim **86** generally has a length and a width greater than a length and a width of the lower base **84**. Additional or alternative configurations of the body **16** are contemplated without departing from the teachings herein.

(23) In various examples, the console assembly **14** includes control buttons **88**, which are generally located at the vehicle-forward second end **82**. The second end **82** may have a recessed region, offset from the remainder of a surface of the upper rim **86** to allow space for the control buttons **88**. The recessed region also lowers the control buttons **88** to provide space for movement of the cover assembly **32** as described herein. The console assembly **14** may also include additional features such as a support surface at the first end **80**, cup holders, etc.
(24) Referring still to FIGS. **1** and **2**, the console assembly **14** defines a storage cavity **90** between the opposing sidewalls **18**, **20**. The storage cavity **90** may be covered by the cover assembly **32** and accessed by moving the cover assembly **32** relative to the upper rim **86** as described further herein.
(25) In the illustrated configuration, the storage cavity **90** is wider at an opening into the storage cavity **90** compared to a bottom of the storage cavity **90**. The wider opening may provide increased access into lower portions of the storage cavity **90**, as well as provide a location for the rails **22**, **24** on the sidewalls **18**, **20** that does not substantially impinge storage space provided by the storage cavity **90**.
(26) Referring still to FIG. **2**, as well as FIG. **3**, the cover assembly **32** is configured to move along multiple movement paths **40**, **42**, **44** relative to the body **16**. The sidewalls **18**, **20** each define the respective rail **22**, **24** along the inner surface **26** thereof proximate to the opening into the storage cavity **90**. The rails **22**, **24** generally extend between the first and second ends **80**, **82** in the fore-aft direction within the vehicle **12** (FIG. **1**). The rails **22**, **24** extend from the inner surface **26** of the sidewalls **18**, **20** into the storage cavity **90**. The rails **22**, **24** may not extend past the inner surfaces **26** of the sidewalls **18**, **20** where the sidewalls **18**, **20** define the narrower width of the storage cavity **90** (e.g., the lower portion). The body **16**, including the rails **22**, **24**, is generally molded from a 30% glass filled polypropylene. The rails **22**, **24** are molded into the sidewalls **18**, **20** and generally extend the entire length of the storage cavity **90** and/or the opening into the storage cavity **90**.
(27) The support bar **28** is configured to extend across the storage cavity **90** and engage each of the rails **22**, **24** defined by the sidewalls **18**, **20**. In this way, the support bar **28** extends in a cross-car direction between the rails **22**, **24**. The support bar **28** is generally constructed of metal or metal alloy materials, such as stainless steel. The support bar **28** is slotted, such that it defines the track **30**, which extends therethrough. The track **30** extends in the cross-car direction between the first and second sidewalls **18**, **20**, extending generally normal to the rails **22**, **24**.
(28) In order to slidably engage the rails **22**, **24**, the support bar **28** is coupled with sliders **100**, **102** at each opposing end. The sliders **100**, **102** are configured to slidably engage with the molded rails **22**, **24**, respectively, and are positioned between the rails **22**, **24** and the support bar **28**. Generally, the engagement between the sliders **100**, **102** and the rails **22**, **24** is a frictional engagement such that a manual force applied to the cover assembly **32** allows the sliders **100**, **102** to translate along the rails **22**, **24** and remain in a selected position when manual force is not being applied. In various aspects, the sliders **100**, **102** are extruded polyoxymethylene (POM).
(29) Referring still to FIGS. **2** and **3**, the cover assembly **32** includes the base panel **34** and the deployable panel **38**. The panels **34**, **38** are operably coupled to the support bar **28**, allowing the panels **34**, **38** to translate in the fore-aft direction with the support bar **28** along the rails **22**, **24** along the first movement path **40**. The connector assembly **36** is configured to couple the base panel **34** to the support bar **28**, as well as allow translation along the track **30** along the second movement path **42** and rotation along the third movement path **44**. The connector assembly **36** generally includes an attachment bracket **110**, a fastener **112**, and a bushing **114**. In various aspects, the base panel **34** is coupled with the support bar **28** via the fastener **112** or a fastener assembly **112**, such as a screw and a washer. The screw and washer are configured to retain the engagement between the base panel **34** and the support bar **28** and still allow movement along the track **30**. The base panel **34** is substantially planar and is generally molded of a composite material, such as a composite material that is about 20% recycled carbon fiber and nylon 6. It is contemplated that

other plastic composites may be utilized without departing from the teachings herein.
(30) The attachment bracket **110** is coupled to a lower surface **116** of the base panel **34**. In non-limiting examples, the attachment bracket **110** may be a zinc die cast attachment bracket **110**. The attachment bracket **110** is generally insert molded to the lower surface **116** of the base panel **34**. In such examples, the attachment bracket **110** is overmolded to the lower surface **116** of the base panel **34** via a covering **118**, which is generally constructed of plastic materials. The attachment bracket **110** includes a shaft **120** that extends away from the base panel **34**. When the base panel **34** is coupled to the body **16** via the support bar **28**, the shaft **120** of the attachment bracket **110** extends through the track **30** toward a bottom of the storage cavity **90**.
(31) The bushing **114** is placed over the shaft **120** of the attachment bracket **110**, allowing the panels **34**, **38** to rotate about a vertical axis **122**. The bushing **114** generally allows rotational movement +/−90 degrees in two directions and, therefore, allows about 180 degrees of rotation overall. The bushing **114** may be a molded POM bushing **114**.
(32) The connector assembly **36** is configured to translate in along the track **30** in response to an applied manual force, generally applied by the user. This allows the cover assembly **32** to translate in the cross-car direction along the second movement path **42**. A frictional interference or engagement between the connector assembly **36** and the support bar **28** allows the movement along the track **30** and retains the cover assembly **32** in the selected position in the absence of the applied force.
(33) Referring still to FIGS. **2** and **3**, the deployable panel **38** of the cover assembly **32** is rotatably coupled to the base panel **34**. The deployable panel **38** has a similar length and width relative to the base panel **34**, though may be different sizes and shapes without departing from the teachings herein. For example, the deployable panel **38** may have an increased thickness compared to the base panel **34**. The deployable panel **38** is generally injection molded from a composite material, which is about 20% recycled carbon fiber and nylon 6. Other plastic composites may be utilized without departing from the teachings herein.
(34) The deployable panel **38** is rotatably coupled to the base panel **34** via hinges **130**. The hinges **130** may be living hinges **130** or hinge assemblies **130** coupled to each of the panels **34**, **38**. Generally, in when the cover assembly **32** is in a center position as illustrated in FIGS. **2** and **3**, the hinges **130** are disposed at vehicle-forward edges of the panels **34**, **38**. The hinges **130** allow the deployable panel **38** to rotate between a folded state and a deployed state as described herein.
(35) A padding **132** may be coupled to the deployable panel **38**. The padding **132** covers one side **134** of the deployable panel **38**, which is an upper side of the deployable panel **38** when in the folded state and a lower side when in the deployed state. Generally, the padding **132** is a polyurethane foam pad, which may be overmolded onto the deployable panel **38** by a cover **136**. The cover **136** may be leather, film, plastic, fabric, vinyl, etc. The cover **136** may extend over the padding **132**, as well as edges of the deployable panel **38**. The padding **132** is generally configured to provide an armrest for occupants of the seating assemblies **64**, **66** (FIG. **1**).
(36) Referring to FIG. **4**, the console assembly **14** is configured to translate along the first movement path **40** defined by the rails **22**, **24**, which is in the fore-aft direction between the first and second ends **80**, **82** of the body **16**. The first movement path **40** may be advantageous for moving the cover assembly **32** to access the storage cavity **90**, as well as for adjusting the cover assembly **32** relative to occupants of the seating assemblies **64**, **66** as described herein. The cover assembly **32** may extend at least partially over the storage cavity **90** as the cover assembly **32** translates and/or is positioned at ends of the rails **22**, **24**. Alternatively, the storage cavity **90** may be free of the cover assembly **32** when the cover assembly **32** is at certain positions such that the opening to the storage cavity **90** is fully accessible (e.g., the cover assembly **32** is moved adjacent to the storage cavity **90** in the fore or aft direction).
(37) Referring still to FIG. **4**, as well as FIG. **5**, the cover assembly **32** is operable between the folded armrest state as illustrated in FIG. **4**, and a deployed or tray state, as illustrated in FIG. **5**.

When in the folded state, the deployable panel **38** is generally positioned over and rests on the base panel **34**. Depending on the configuration of the deployable panel **38**, the deployable panel **38** generally covers an entire upper surface **138** of the base panel **34**. When in the folded state and when the connector assembly **36** is in a center of the track **30** on the support bar **28**, the panels **34**, **38** generally extend from the first sidewall **18** to the second sidewall **20** over the storage cavity **90**. Further, when in the center position, the folded state, and in the center of the track **30**, side edges of the panels **34**, **38** align with the rails **22**, **24**.

(38) When in the deployed state, the deployable panel **38** rotates about 180 degrees about a horizontal axis **150** defined by the hinges **130**. In this way, when in the deployed state, the deployable panel **38** is disposed adjacent to the base panel **34**. Once in the deployed state, the deployable panel **38** has a surface **152** that aligns with the upper surface **138** of the base panel **34** to form a tray or table surface. The hinges **130** are generally positioned between or below the tray surface defined by the adjacent panels **34**, **38** so as not to impinge on the generally coplanar or flat table or tray surface defined by the deployed cover assembly **32**. The padding **132** disposed on the deployable panel **38** is then on the lower side of the deployable panel **38**.

(39) Referring still to FIGS. **4** and **5**, as well as FIG. **6**, the cover assembly **32** is operable between the center position where the hinges **130** are at the vehicle-forward or console-forward edges of the panels **34**, **38**, as illustrated in FIG. **4**, the first side position, as illustrated in FIG. **5**, and the second side position, as illustrated in FIG. **6**. In each of the center position, the first side position, and the second side position, the cover assembly **32** may be in either of the folded state or the deployed state and may be disposed at any position along the rails **22**, **24** and/or the track **30**. In this way, the cover assembly **32** may rotate to either of the side positions to provide a different configuration for the armrest or to provide the tray surface for different occupants.

(40) The cover assembly **32** is configured to translate along the third movement path **44** as the cover assembly **32** rotates between the center position and the side positions. The cover assembly **32** is configured to rotate about the vertical axis **122** while the panels **34**, **38** translate along a horizontal plane. The cover assembly **32** is configured to rotate up to about 90 degrees in the first direction from the center position, where the hinges **130** are at vehicle-forward edges, to the first side position, where the hinges **130** are disposed proximate the first sidewall **18** and the first seating assembly **64** (e.g., the driver seat **64**). The first side position may be any position between the center position, considered about zero degrees of rotation, and the full rotation of about 90 degrees in the first direction. The cover assembly **32** may be utilized in the folded state or may be adjusted to the deployed state when in the first side position.

(41) When in the deployed state in the first side position, the base panel **34** extends at least partially over the storage cavity **90** and the deployable panel **38** extends at least partially over the first seating assembly **64**. Depending on the position of the connector assembly **36** relative to the track **30**, the deployable panel **38** may extend partially over the storage cavity **90** or may be disposed adjacent to an outer surface of the first sidewall **18** adjacent to the body **16**. In the deployed state in the first side position, the cover assembly **32** provides the tray or table surface primarily or fully for the occupant in the first seating assembly **64**. While the first seating assembly **64** is illustrated as the driver seat **64**, it is contemplated that the deployed state of the cover assembly **32** is utilized when the vehicle **12** is in a stationary condition. However, it is anticipated that technology or the regulatory framework may evolve in the future to where using the cover assembly **32** as a tray or table in a moving vehicle **12** becomes safe and permissible. Accordingly, the use and function of the cover assembly **32** may align with any current technology and regulatory framework.

(42) Referring still to FIG. **6**, the cover assembly **32** is configured to rotate up to about 90 degrees in the second direction from the center position to the second side position, where the hinges **130** are disposed proximate to the second sidewall **20** and the second seating assembly **66** (e.g., the passenger seat **66**). The second side position may be any position between the center position, considered about zero degrees of rotation, and the full rotation of about 90 degrees in the second

direction. Similar to the first side position (FIG. **5**), when in the second side position, the cover assembly **32** may be utilized in the folded state to provide a different configuration for the armrest and in the deployed state to provide the tray surface primarily or fully for the occupant of the second seating assembly **66**. The base panel **34** extends at least partially over the storage cavity **90** and the deployable panel **38** extends at least partially over the second seating assembly **66**, depending on the position of the connector assembly **36** relative to the track **30**.

(43) Referring to FIG. **7**, the cover assembly **32** is also configured to translate along the second movement path **42** along the track **30** when in the first and second side positions. While the cover assembly **32** is illustrated in FIG. **7** in the first side position, it is understood that the description herein is also applicable for when the cover assembly **32** is in the second side position. Whether in the folded state or the deployed state, the cover assembly **32** is configured to move in the cross-car direction along the track **30** of the support bar **28**. This allows the tray surface to be more fully positioned over the first seating assembly **64** as well as the occupant in the first seating assembly **64**.

(44) Additionally, when in the first side position and in the deployed state, the cover assembly **32** being adjusted along the track **30** toward the second seating assembly **66** may provide a more central tray or table surface that may be utilized by occupants in either or both of the seating assemblies **64**, **66**. This central tray or table surface may also be available by adjusting the cover assembly **32** to the second side position in the deployed state and moving the cover assembly **32** along the track **30** toward the first seating assembly **64**.

(45) Referring to FIGS. **4-7**, the cover assembly **32** is configured to translate along the three movement paths **40**, **42**, **44**, which are generally defined along the same horizontal plane (i.e., a single plane). In this way, the cover assembly **32** may translate fore-and-aft, translate in the cross-car direction, and rotate between the side positions to provide a variety of positions for the cover assembly **32**. The cover assembly **32** may be moved along any of the movement paths **40**, **42**, **44** in combination with one another. Accordingly, the cover assembly **32** can be moved along the rails **22**, **24**, along the track **30**, rotated about the vertical axis **122**, and combinations thereof. Further, while moving along any of the movement paths **40**, **42**, **44**, the cover assembly **32** may be in the folded state or the deployed state. This configuration provides customizable positioning of the cover assembly **32** for both the folded cover assembly **32** providing the armrest and the deployed cover assembly **32** providing the tray surface.

(46) Referring to FIGS. **8-10**, the cover assembly **32** may have a variety of configurations that each may translate along the first, second, and third movement paths **40**, **42**, **44**. For example, as illustrated in FIGS. **8** and **9**, the cover assembly **32** may include a single elongated panel **160**. The elongated panel **160** may have tapering ends, which may be advantageous during rotation of the elongated panel **160** along the third movement path **44** while occupants are in the seating assemblies **64**, **66**. A longitudinal extent of the elongated panel **160** may extend in the fore-aft direction to substantially cover the storage cavity **90** and may extend in the cross-car direction to provide the tray surface. The elongated panel **160** may extend between the first and second ends **80**, **82** to cover the storage cavity **90** and may rotate to provide the tray or table.

(47) Referring to FIG. **10**, in an additional or alternative configuration, cover assembly **32** may include three panels **162**, **164**, **166**, including a connector panel **162** and two opposing side panels **164**, **166**. In this way, the two side panels **164**, **166** are configured to adjust to the folded state over the connector panel **162** and adjacent to one another and deploy to be on opposing sides of the connector panel **162** adjacent to the connector panel **162**. Other configurations of the cover assembly **32** that translates along three movement paths **40**, **42**, **44** are contemplated without departing from the teachings herein.

(48) Referring to FIGS. **1-10**, the cover assembly **32** is configured to move along the three movement paths **40**, **42**, **44** when in the folded state and the deployed state. The occupants in the seating assemblies **64**, **66** may adjust the cover assembly **32** to any position along each of the three

movement paths **40**, **42**, **44**, separately or in combinations, to provide a personalized position of the cover assembly **32**. This is advantageous for providing the armrest, as well as the tray surface for the occupants in one or both seating assemblies **64**, **66** in accordance with current technology and regulatory framework. The cover assembly **32** may be moved along any of the movement paths **40**, **42**, **44** in response to manual force applied by the occupants in the vehicle **12**. The engagements between the sliders **100**, **102** and the rails **22**, **24**, the connector assembly **36** and the track **30**, and the support bar **28**, the bearing, and the shaft **120** of the attachment bracket **110** are frictional engagements that allow the movement of the components relative to one another and retain the components in the select position when the manual force is not being applied.

(49) Referring to FIG. **11**, as well as FIG. **1-10**, the console assembly **14** is generally part of the console system **10** for the vehicle **12**. The console system **10** provides different features used in combination with the console assembly **14**. The vehicle **12** includes a controller **170**, which is generally an electronic control unit for the vehicle **12**. The controller **170** includes a processor **172**, a memory **174**, and other control circuitry. Instructions or routines **176** are stored within the memory **174** and executable by the processor **172**. The controller **170** is in communication with a variety of features within the vehicle **12**, which may be controlled (e.g., activated, deactivated, adjusted, etc.) in response to the state and/or position of the cover assembly **32**.

(50) For example, the vehicle **12** may include an interior imager **180**. The interior imager **180** may be a charge coupled device, a complementary metal oxide semiconductor, or any type of camera. The interior imager **180** defines a field of view **182** that includes the console assembly **14**. The field of view **182** may be sized and oriented to include the cover assembly **32** in each state and position. Additionally or alternatively, the interior imager **180** may have lenses or similar features that adjust the field of view **182**. The interior imager **180** is configured to capture image data from the field of view **182** and communicate the image data to the controller **170**.

(51) The controller **170** generally includes image processing routines **176**, which are configured to process and analyze the image data. Based on the image data, the controller **170** is configured to determine whether the cover assembly **32** is in the folded state or the deployed state, as well as the position of the cover assembly **32** along the movement paths **40**, **42**, **44**. In this way, the controller **170** is configured to determine whether the cover assembly **32** moved in the vehicle-forward direction, is in one of the side positions, is positioned centrally or over one of the seating assemblies **64**, **66**, etc.

(52) Referring still to FIG. **11**, the controller **170** is generally configured to adjust functions of the vehicle **12** in response to the position or state of the cover assembly **32**. For example, when the cover assembly **32** is in the deployed state, the controller **170** may prevent the vehicle **12** from being activated or shifted into a drive state. In this way, the controller **170** is configured to "lockout" the vehicle **12** in response to the state of the cover assembly **32**. The vehicle **12** may be "locked out" of the drive state when the cover assembly **32** is in the deployed state regardless of the position of the cover assembly **32** (e.g., first side position v. second side position).

(53) Additionally or alternatively, the controller **170** may "lockout" the vehicle **12** from the drive state when the state and position of the cover assembly **32** are indicative of the occupant in the driver seat **64** using the tray surface. In this way, when the cover assembly **32** is in the deployed state and in the first side position, the vehicle **12** may not be activated or adjusted into the drive state. This may also be true when the cover assembly **32** is in the second side position and shifted along the track **30** toward the first seating assembly **64** to provide the central tray surface. However, when the cover assembly **32** is in the deployed state and the second side position, extending over the second seating assembly **66**, the vehicle **12** may be adjusted to the drive state. This configuration allows the occupant in the passenger seat **66** to use the tray surface in the moving vehicle **12**, but not the occupant in the driver seat **64**.

(54) The conditions for when the controller **170** "locks" the vehicle **12** out of the drive state may be based on current technology and regulatory framework. It is contemplated that in the future

technology and/or the regulatory framework may permit the use of the tray or table surface by an occupant in the first seating assembly **64** and/or the second seating assembly **66** while the vehicle **12** is in the driving state or moving. In such scenarios, the cover assembly **32** may be used in accordance with the current technology and regulatory framework.

(55) Referring still to FIG. **11**, the vehicle **12** generally includes multiple light sources **184** configured to illuminate areas around the console assembly **14** and/or the seating assemblies **64**, **66**. In the illustrated example, the light sources **184** include a driver map light **186** configured to emit light over the first seating assembly **64**, a passenger map light **188** configured to emit light over the second seating assembly **66**, and a submarine or overhead console light **190** configured to emit light over the console assembly **14**.

(56) In various examples, the light sources **184** are activated based on the position and/or state of the cover assembly **32**. For example, when the cover assembly **32** is in the deployed state and the first side position, extending over the first seating assembly **64**, the controller **170** is configured to activate the driver map light **186** to provide light on the tray surface over the first seating assembly **64**. When the cover assembly **32** is in the deployed state and in the second side position, extending over the second seating assembly **66**, the controller **170** is configured to activate the passenger map light **188** to provide light to the tray surface over the second seating assembly **66**. When the cover assembly **32** is in the deployed state and providing a central tray surface (e.g., in the first side position and disposed in the track **30** proximate the second seating assembly **66** or in the second side position and disposed in the track **30** proximate the first seating assembly **64**), the controller **170** is configured to activate the overhead console light **190** to provide lighting for the central tray surface. The light sources **184** may be activated at different intensities based on the type of light sources **184**, the time of day, the available light determined by the interior imager **180** or other light sensors, etc.

(57) With further reference to FIG. **11**, the controller **170** may be configured to adjust a position of one or both of the seating assemblies **64**, **66** in response to the state and/or position of the cover assembly **32**. Each of the seating assemblies **64**, **66** includes an actuator **200**, **202** or multiple actuators **200**, **202** configured to adjust the respective seating assembly **64**, **66**. The actuators **200**, **202** are generally configured to adjust the respective seating assembly **64**, **66** in the fore-aft direction, vertically, angle of a seat base, and/or angle of a seat back. When the controller **170** determines that the cover assembly **32** is in the deployed state providing the central tray surface, the controller **170** may be configured to adjust one or both of the seating assemblies **64**, **66** forward and/or upright to a task orientation or work position, providing better access to the center tray surface. When the cover assembly **32** is in the deployed state in the first side position, the controller **170** may be configured to adjust the first seating assembly **64** forward and upright to provide the task orientation. Similarly, for the second seating assembly **66**, when the cover assembly **32** is in the second side position and the deployed state, the second seating assembly **66** may be adjusted forward and upright to the task orientation. Accordingly, the controller **170** may automatically adjust the seating assembly **64**, **66** for the occupant who is using the tray surface to provide better or increased access to the tray surface.

(58) According to various aspects, the vehicle **12** may include one or both of an exterior imager **208** and a radar sensor **210**. Each of the exterior imager **208** and the radar sensor **210** may be coupled to a vehicle-forward portion of the vehicle **12** and have a field of view extending in a vehicle-forward direction to capture data and communicate the data to the controller **170**. Using the data from one or both of the exterior imager **208** and the radar sensor **210**, the controller **170** may be configured to determine whether objects, such as a people or other vehicles **12**, are approaching the vehicle **12**, the distance of the objects, and/or the speed of the objects.

(59) The controller **170** is in communication with a display **216** of the center stack **70**. The controller **170** may be configured to display the information from the exterior imager **208** and/or the radar sensor **210** on the display **216** for the occupant in the interior compartment **60** to view.

The displayed information may be received data, such as image data, and/or processed information, such as a notification of a person detected within the field of view and at what distance is the person. This may be advantageous for the occupant utilizing the cover assembly **32** in the deployed state as the tray to notify the occupant of an approaching person. For example, the occupant working at a mobile workstation provided by the tray or eating lunch may desire to be alerted when someone is approaching that may wish to speak to the occupant.
(60) Referring again to FIG. **1** and to FIG. **11**, when the cover assembly **32** is moved toward the vehicle-forward second end **82** of the console assembly **14**, whether in the folded state or the deployed state, the cover assembly **32** may extend over the control buttons **88** on the upper rim **86** of the body **16**. When the controller **170** determines that the console assembly **14** is positioned over the control buttons **88** and, therefore, likely impeding access to the control buttons **88**, the controller **170** is configured to display selectable control features **218** on the display **216** that correspond with the covered control buttons **88**. For example, if the control buttons **88** adjust heating in the seating assemblies **64**, **66**, the control features **218** on the display **216** are configured to adjust the heating in the seating assemblies **64**, **66**. When the cover assembly **32** is moved in the vehicle-rearward direction, uncovering the control buttons **88**, the controller **170** may remove the control features **218** from the display **216**. In this way, the user does not have to readjust the cover assembly **32** to control certain features and the controls remain readily accessible.
(61) Referring to FIGS. **1-11**, the cover assembly **32** is configured to be utilized in the folded state as the armrest and the deployed state as the table or tray. The cover assembly **32** is configured to translate along the first movement path **40** in the fore-aft direction, which may occur while the cover assembly **32** is in the folded state or the deployed state, as well as in the center position and the side positions. In this way, the deployed tray over one of the seating assemblies **64**, **66** may be adjusted for the position preference of the occupant. The cover assembly **32** is also configured to translate along the second movement path **42** in the cross-car direction. The cover assembly **32** may move along the second movement path **42** when in the folded state or the deployed state, as well as in the center position and the side positions. Further, the cover assembly **32** is configured to rotate about the vertical axis **122** and move the cover assembly **32** along the third movement path **44** between the first side position and about 180 degrees to the second side position. The cover assembly **32** is configured to translate along the same horizontal plane as the cover assembly **32** moves along the first movement path **40**, the second movement path **42**, and the third movement path **44**.
(62) The cover assembly **32** may provide a mobile workstation, which may be advantageous for providing space for an occupant of the vehicle **12** to work or provide a place to eat lunch while at a worksite while the vehicle **12** is parked and stationary. The cover assembly **32** is utilized within the current technology and regulatory framework, which in the future could include the use of the cover assembly **32** as the tray in the moving vehicle **12** for one or both of the occupants in the seating assemblies **64**, **66**. The cover assembly **32** is also configured to be used in conjunction with other features of the vehicle **12** as described herein to maximize the user experience with the cover assembly **32**.
(63) The vehicle **12** is a wheeled motor vehicle **12**, which may be a sedan, a sport-utility vehicle, a truck, a van, a crossover, and/or other styles of vehicle **12**. The vehicle **12** may be a manually operated vehicle **12** (e.g., operated with a human driver), a fully autonomous vehicle **12** (e.g., operated without a human driver), or a partially autonomous vehicle **12** (e.g., operated with or without a human driver). Additionally, the vehicle **12** would be utilized for personal and/or commercial purposes, such as ride providing services (e.g., chauffeuring) and/or ridesharing services.
(64) Use of the present device may provide for a variety of advantages. For example, the cover assembly **32** may be utilized to cover and uncover the storage cavity **90** in the console assembly **14**. Additionally, the cover assembly **32** is configured to translate along three different movement paths

40, **42**, **44** to provide a variety of positions and orientations of the cover assembly **32** relative to the body **16**, as well as the occupants in the adjacent seating assemblies **64**, **66**. Also, the cover assembly **32** is configured to translate fore-and-aft, translate in the cross-car direction, and rotate. Further, the cover assembly **32** is configured to be utilized in the folded state as the armrest and in the deployed state to provide the table or tray surface. Moreover, the console assembly **14** and cover assembly **32** may provide a mobile workstation that may be utilized within current technological and regulatory frameworks. Further, the console system **10** maximizes the user experience with the cover assembly **32** by adjusting lighting, seating position, and displayed information on the display **216** when the cover assembly **32** is in select states or positions. Additional benefits or advantages may be realized and/or achieved.

(65) The device disclosed herein is further summarized in the following paragraphs and is further characterized by combinations of any and all of the various aspects described therein.

(66) According to various examples, a vehicle console system includes a body having first and second sidewalls. Each of the first and second sidewalls defines a rail along an inner surface thereof. A support bar slidably engages the rails. The support bar defines a track. A cover assembly is operably coupled to the support bar. The cover assembly includes a base panel coupled to the support bar via a connector assembly and a deployable panel rotatably coupled to the base panel. The cover assembly is configured to translate along a first movement path with the support bar along the rails and a second movement path along the track of the support bar. The cover assembly is configured to rotate along a third movement path between a first side position and a second side position. Embodiments of the present disclosure may include one or a combination of the following features: the deployable panel is operable between a folded armrest state positioned on an upper surface of the base panel and a deployed tray state adjacent to the base panel; the cover assembly is configured to rotate 90 degrees in a first direction from a center position to the first side position and 90 degrees in a second direction from a center position to the second side position; the deployable panel is coupled to the base panel via hinges; the hinges are disposed in a console-forward direction when the cover assembly is in the center position; an interior imager defining a field of view including the cover assembly; the interior imager is configured to capture image data within the field of view; a controller communicatively coupled to the interior imager; the controller is configured to determine when the cover assembly is in a deployed tray state based on the image data; a light source communicatively coupled to the controller; the controller is configured to activate the light source when the cover assembly is in the deployed tray state; a display communicatively coupled to the controller; the console assembly includes control buttons at an end thereof; and selectable control features are configured to be displayed on the display when the cover assembly is positioned over the control buttons.

(67) According to various examples, a vehicle console assembly includes a body defining a storage cavity between first and second sidewalls. A support bar coupled to each of the first and second sidewalls. The support bar defines a track extending between the first and second sidewalls. A cover assembly is operably coupled to the support bar. The cover assembly includes a first panel. A second panel is rotatably coupled to the first panel. The second panel is operable between a folded state and a deployed state. A connector assembly is coupled to a lower surface of the first panel and extends through the track of the support bar. The cover assembly is configured to rotate in a first direction from a center position to a first side position and in a second direction from the center position to a second side position. The cover assembly is configured to translate along the track. Embodiments of the present disclosure may include one or a combination of the following features: the first and second sidewalls each define a rail; the support bar is configured to translate along the rails; the cover assembly is configured to translate along a first movement path defined by the rails and a second movement path defined by the track, the second movement path being normal to the first movement path; the cover assembly is configured to translate along a single plane when translating along the first movement path, the second movement path, and when rotating between

the first and second side positions; the cover assembly is configured to rotate about a vertical axis between the first side position, the center position, and the second side position; the cover assembly includes padding coupled to the second panel to provide an armrest when the second panel is in the folded state; the first panel is disposed at least partially over the storage cavity when the cover assembly is in the folded state and the deployed state; the second panel is disposed proximate the first sidewall when the cover assembly is in the deployed state in the first side position; the second panel is disposed proximate the second sidewall when the cover assembly is in the deployed state in the second side position; and a surface of the second panel is coplanar with an upper surface of the first panel when the cover assembly is in the deployed state to form a tray.
(68) According to various examples, a console assembly for a vehicle includes a body having sidewalls. Each sidewall defines a rail. A support bar slidably engages the rails. The support bar defines a track. A cover assembly is operably coupled to the support bar. A connector assembly is coupled to the cover assembly and engages the support bar. The cover assembly is configured to translate along a first movement path with the support bar via the rails, translate along a second movement path along the track via the connector assembly, and rotate about a vertical axis along a third movement path via the connector assembly. Embodiments of the present disclosure may include one or a combination of the following features: the cover assembly includes a deployable panel rotatably coupled to a base panel; the deployable panel is operable between a folded state over the base panel and a deployed state adjacent to the base panel; the cover assembly is configured to rotate 180 degrees when translating along the third movement path between a first side position and a second side position; and the first movement path is normal to the second movement path.
(69) For purposes of this disclosure, the term "coupled" (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.
(70) It should be noted that the sensor examples discussed above might include computer hardware, software, firmware, or any combination thereof to perform at least a portion of their functions. For example, a sensor may include computer code configured to be executed in one or more processors and may include hardware logic/electrical circuitry controlled by the computer code. These example devices are provided herein for purposes of illustration and are not intended to be limiting. Examples of the present disclosure may be implemented in further types of devices, as would be known to persons skilled in the relevant art(s).
(71) The various illustrative logical blocks, modules, controllers, and circuits described in connection with the embodiments disclosed herein may be implemented or performed with application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), general purpose processors, digital signal processors (DSPs) or other logic devices, discrete gates or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be any conventional processor, controller, microcontroller, state machine or the like. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.
(72) It is also important to note that the construction and arrangement of the elements of the invention as shown in the exemplary examples is illustrative only. Although only a few examples of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations

in sizes, dimensions, structures, shapes, and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connectors or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system might be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary examples without departing from the spirit of the present innovations.
(73) Modifications of the disclosure will occur to those skilled in the art and to those who make or use the disclosure. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the disclosure, which is defined by the following claims, as interpreted according to the principles of patent law, including the doctrine of equivalents.
(74) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle console system, comprising: a body having first and second sidewalls, wherein each of the first and second sidewalls defines a rail, respectively, along an inner surface thereof; a support bar slidably engaging the rails, wherein the support bar defines a track; a cover assembly operably coupled to the support bar, wherein the cover assembly includes: a base panel coupled to the support bar via a connector assembly; and a deployable panel rotatably coupled to the base panel and operable between a folded armrest state positioned on an upper surface of the base panel and a deployed tray state adjacent to the base panel, wherein the cover assembly is configured to translate along a first movement path with the support bar along the rails and a second movement path along the track of the support bar, wherein the cover assembly is configured to rotate along a third movement path between a first side position and a second side position, and wherein when the deployable panel is in the deployed tray state, the base panel extends at least partially over the body and the deployable panel extends at least partially over a first seating assembly when in the first side position and extends at least partially over a second seating assembly when in the second side position.
2. The vehicle console system of claim 1, wherein the cover assembly is configured to rotate 90 degrees in a first direction from a center position to the first side position and 90 degrees in a second direction from the center position to the second side position.
3. The vehicle console system of claim 2, wherein the deployable panel is coupled to the base panel via hinges, and wherein the hinges are disposed in a console-forward direction when the cover assembly is in the center position.
4. The vehicle console system of claim 1, further comprising: an interior imager defining a field of view including the cover assembly, wherein the interior imager is configured to capture image data within the field of view; and a controller communicatively coupled to the interior imager, wherein the controller is configured to determine when the cover assembly is in the deployed tray state

based on the image data.
5. The vehicle console system of claim 4, further comprising: a light source communicatively coupled to the controller, wherein the controller is configured to activate the light source when the cover assembly is in the deployed tray state.
6. The vehicle console system of claim 4, further comprising: a display communicatively coupled to the controller, wherein the console assembly includes control buttons at an end thereof, and wherein selectable control features are configured to be displayed on the display in response to the cover assembly being positioned over the control buttons.
7. The vehicle console system of claim 4, wherein the first seating assembly has a seat actuator, the seat actuator being in communication with the controller, and wherein the seat actuator is configured to adjust the first seating assembly to a task orientation when the cover assembly is in the deployed tray state.
8. A vehicle console assembly, comprising: a body defining a storage cavity between first and second sidewalls; a support bar coupled to each of the first and second sidewalls, wherein the support bar defines a track extending between the first and second sidewalls; and a cover assembly operably coupled to the support bar, wherein the cover assembly includes: a first panel; a second panel rotatably coupled to the first panel, wherein the second panel is operable between a folded state and a deployed state; and a connector assembly coupled to a lower surface of the first panel and extending through the track of the support bar, wherein the cover assembly is configured to rotate in a first direction from a center position to a first side position and in a second direction from the center position to a second side position, and wherein the cover assembly is configured to translate along the track.
9. The vehicle console assembly of claim 8, wherein the first and second sidewalls each define a rail, and wherein the support bar is configured to translate along the rails, and wherein the cover assembly is configured to translate along a first movement path defined by the rails and a second movement path defined by the track, the second movement path being normal to the first movement path.
10. The vehicle console assembly of claim 9, wherein the cover assembly is configured to translate along a single plane when translating along the first movement path, the second movement path, and when rotating between the first and second side positions.
11. The vehicle console assembly of claim 8, wherein the cover assembly is configured to rotate about a vertical axis between the first side position, the center position, and the second side position.
12. The vehicle console assembly of claim 8, wherein the cover assembly includes padding coupled to the second panel to provide an armrest when the second panel is in the folded state.
13. The vehicle console assembly of claim 8, wherein the first panel is disposed at least partially over the storage cavity when the cover assembly is in the folded state and the deployed state.
14. The vehicle console assembly of claim 8, wherein the second panel is disposed proximate the first sidewall when the cover assembly is in the deployed state in the first side position, and wherein the second panel is disposed proximate the second sidewall when the cover assembly is in the deployed state in the second side position.
15. The vehicle console assembly of claim 8, wherein a surface of the second panel is coplanar with an upper surface of the first panel when the cover assembly is in the deployed state to form a tray.
16. A console assembly for a vehicle, comprising: a body having sidewalls, wherein each sidewall defines a rail; a support bar slidably engaging the rails, wherein the support bar defines a track; a cover assembly operably coupled to the support bar; and a connector assembly coupled to a lower surface of the cover assembly and extending into the track of the support bar, wherein the cover assembly is configured to translate along a first movement path with the support bar via the rails, translate along a second movement path along the track via the connector assembly, and rotate

about a vertical axis along a third movement path via the connector assembly.
17. The console assembly of claim 16, wherein the first movement path is normal to the second movement path.
18. The console assembly of claim 6, wherein the selectable control features correspond with the covered control buttons.
19. The console assembly of claim 16, wherein the rails are elongated slots extending along inner surfaces of the sidewalls, respectively, wherein the support bar extends into each of the rails, and wherein the cover assembly is disposed above each of the rails.
20. The console assembly of claim 16, wherein the connector assembly includes a shaft extending through the track of the support bar, and wherein a bushing is placed over the shaft.